

Mario Gutiérrez

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WORK EXPERIENCE

Universidad Politécnica de Madrid

Nov. 2024 – Present

Research Intern

Madrid, Spain

- Collaborated on the **architectural design** and development of a risk assessment framework for the **Future Combat Air System (FCAS)** defense project.
 - Developed a **cyberthreat dashboard** to deliver data-driven risk metrics for strategic decision-making.
 - Designed a **dynamic risk assessment framework** using ontology-based modeling; engineered logic to automatically propagate and measure risk across interconnected asset networks.

EDUCATION

Universidad Politécnica de Madrid

June, 2026

BSc. Data Engineering and Systems

Madrid, Spain

- Highest Academic Record in the First Year.
- 3.3/4.0 GPA
- Skills: Big Data, Databases (SQL, NoSQL), Distributed Systems, Programming (Python, Scala), Cloud Computing, Data Networks, Computer Architecture, Operating Systems, Cybersecurity, Probability Theory, Information Theory, Linear Algebra, Machine Learning, Optimization, Signal Processing, Programming (C++, C), React...

Universidad Politécnica de Madrid

June, 2027

MSc. Data Science

Madrid, Spain

- Innovation & Entrepreneurship Minor (European Institute of Innovation & Technology)

KTH Royal Institute of Technology

June, 2028

MSc. Data Science

Stockholm, Sweden

HONORS & AWARDS

II Edition IndesIAHack Winner

November, 2025

IndesIA

Madrid, Spain

- Developed a **sustainable AI routing system** designed to redirect queries to the **optimal model** by balancing accuracy and cost; implemented a classification logic based on topic and difficulty, utilizing empirical model benchmarks to automate selection.

SIDE PROJECTS

mini-gpt

April, 2026

- Built a **GPT-2 architecture from scratch** in **PyTorch** and trained it on the **TinyStories dataset**. The result is a **lightweight model** that generates **coherent, short form children's stories**.

Data Orchestrator

November, 2025

- Automated **taxi data ingestion** from CSVs into **Elasticsearch** and **MongoDB**; built a **Flask** interface with a **Rasa chatbot** using dual-path logic to route queries between static files and dynamic database lookups.